Exploration and Skill Learning

CS 185/285

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Recap: what's the problem?

this is easy (mostly)

Why?

this is impossible

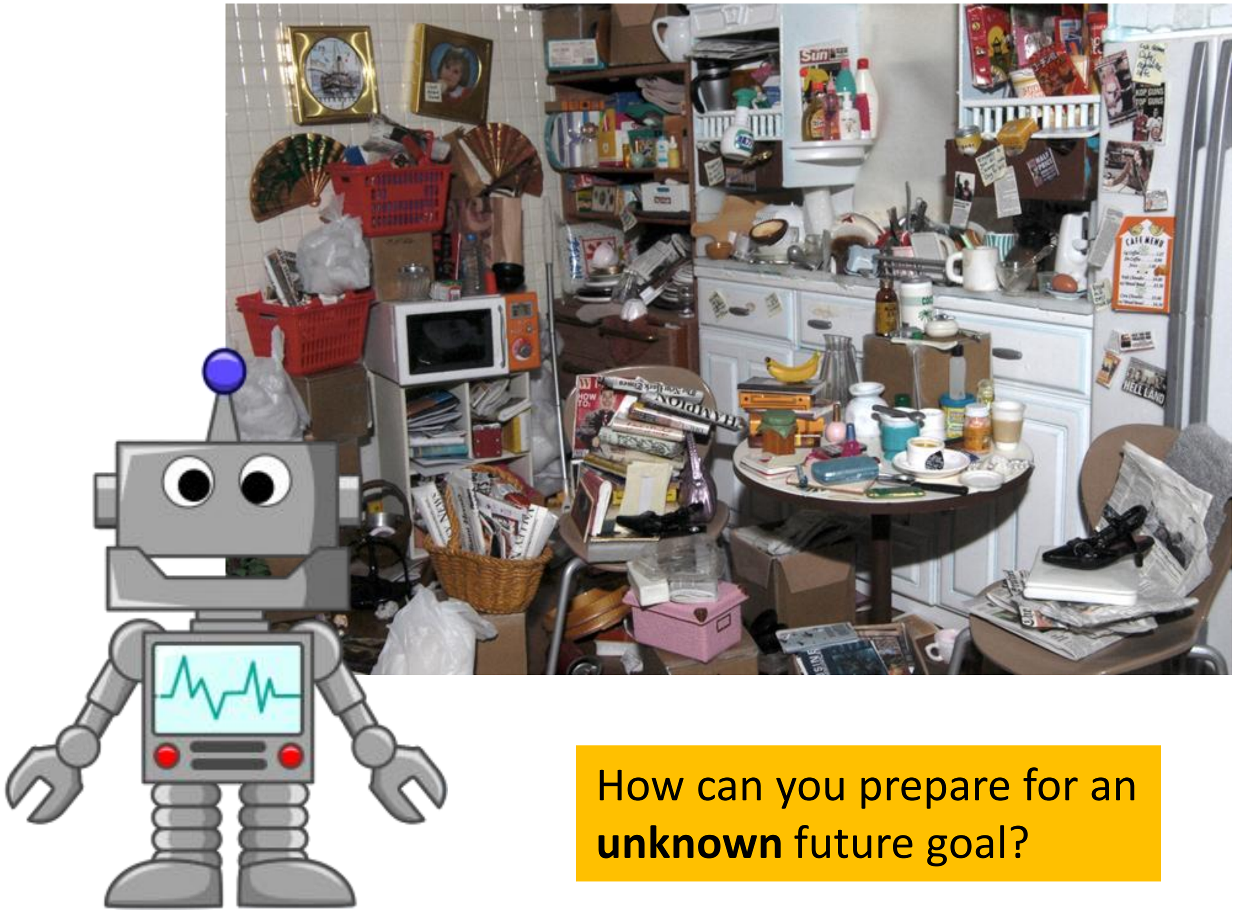
Unsupervised learning of diverse behaviors

What if we want to recover diverse behavior **without any reward function at all**?

Why?

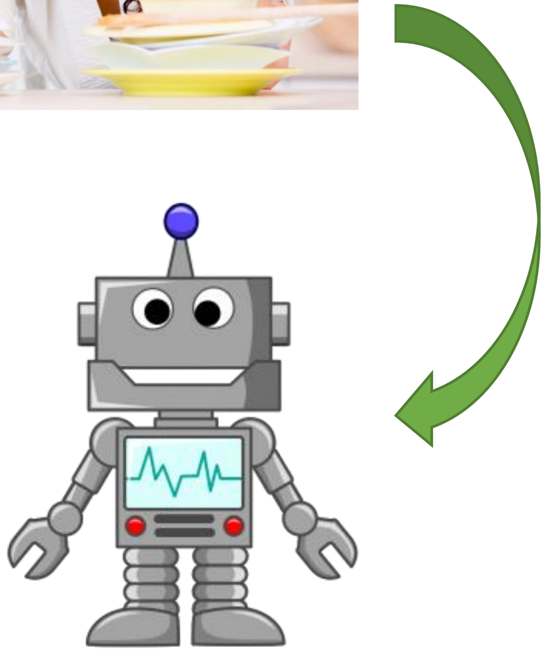
- *Learn skills without supervision, then use them to accomplish goals*
- *Learn sub-skills to use with hierarchical reinforcement learning*
- *Explore the space of possible behaviors*

An Example Scenario



How can you prepare for an **unknown** future goal?

training time: unsupervised



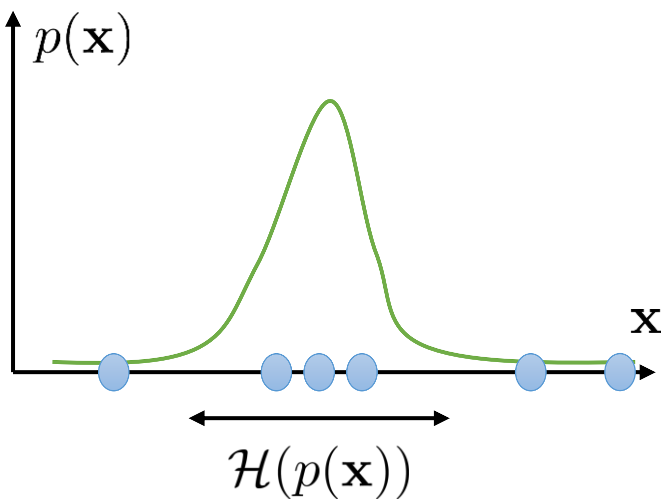
Part 1:
Information theory recap

Some useful identities

$p(\mathbf{x})$ distribution (e.g., over observations $\mathbf{x}$)

$$\mathcal{H}(p(\mathbf{x})) = -E_{\mathbf{x}\sim p(\mathbf{x})}[\log p(\mathbf{x})]$$

entropy – how "broad" $p(\mathbf{x})$ is



Some useful identities

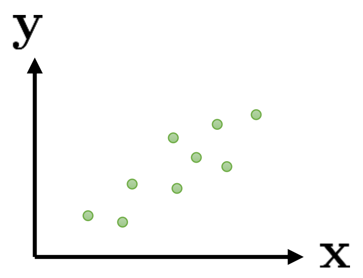
entropy – how "broad" $p(\mathbf{x})$ is

$$\mathcal{H}(p(\mathbf{x})) = -E_{\mathbf{x}\sim p(\mathbf{x})}[\log p(\mathbf{x})]$$

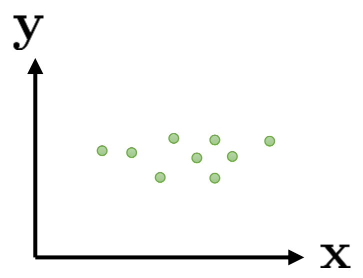
$$\mathcal{I}(\mathbf{x};\mathbf{y}) = D_{\mathrm{KL}}(p(\mathbf{x},\mathbf{y})\|p(\mathbf{x})p(\mathbf{y}))$$

$$= E_{(\mathbf{x},\mathbf{y})\sim p(\mathbf{x},\mathbf{y})}\left[\log \frac{p(\mathbf{x},\mathbf{y})}{p(\mathbf{x})p(\mathbf{y})}\right]$$

$$= \mathcal{H}(p(\mathbf{y})) - \mathcal{H}(p(\mathbf{y}|\mathbf{x}))$$



high MI: $\mathbf{x}$ and $\mathbf{y}$ are *dependent*



low MI: $\mathbf{x}$ and $\mathbf{y}$ are *independent*

Information theoretic quantities in RL

$\pi(\mathbf{s})$ state *marginal* distribution of policy π

quantifies *coverage*

$\mathcal{H}(\pi(\mathbf{s}))$ state *marginal* entropy of policy π

example of mutual information: "empowerment" (Polani et al.)

$$\mathcal{I}(\mathbf{a}_t; \mathbf{s}_{t+1}) = \mathcal{H}(\mathbf{s}_{t+1}) - \mathcal{H}(\mathbf{s}_{t+1} | \mathbf{a}_t)$$

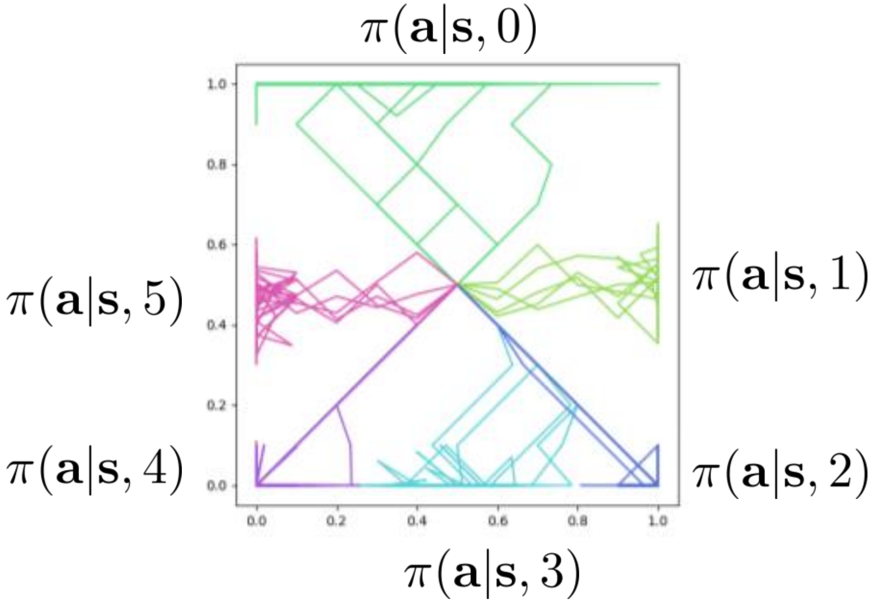
can be viewed as quantifying "control authority" in an information-theoretic way

Part 2:
Learning skills with empowerment

Learning diverse skills

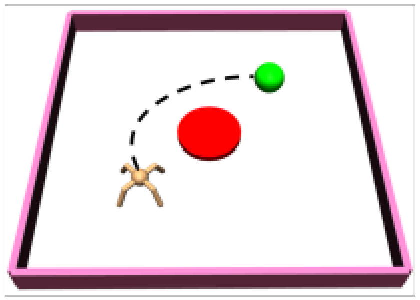
$$\pi(\mathbf{a}|\mathbf{s}, z)$$

task index



Reaching diverse **goals** is not the same as performing diverse **tasks**

not all behaviors can be captured by **goal-reaching**



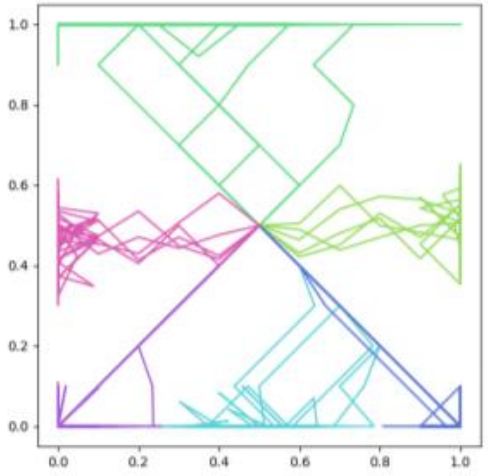
Intuition: different **skills** should visit different **state-space regions**

Eysenbach, Gupta, Ibarz, Levine. **Diversity is All You Need.**

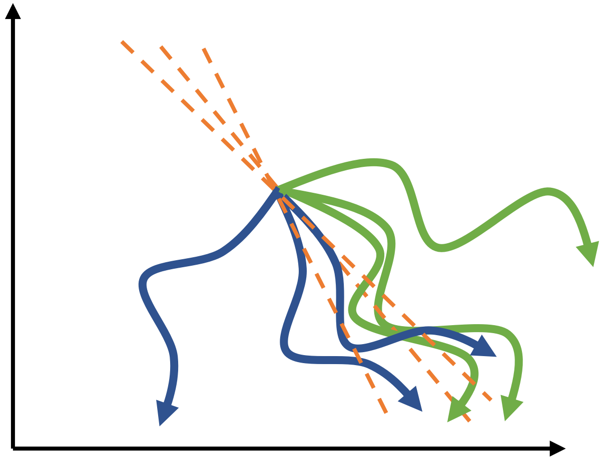
Diversity-promoting reward function

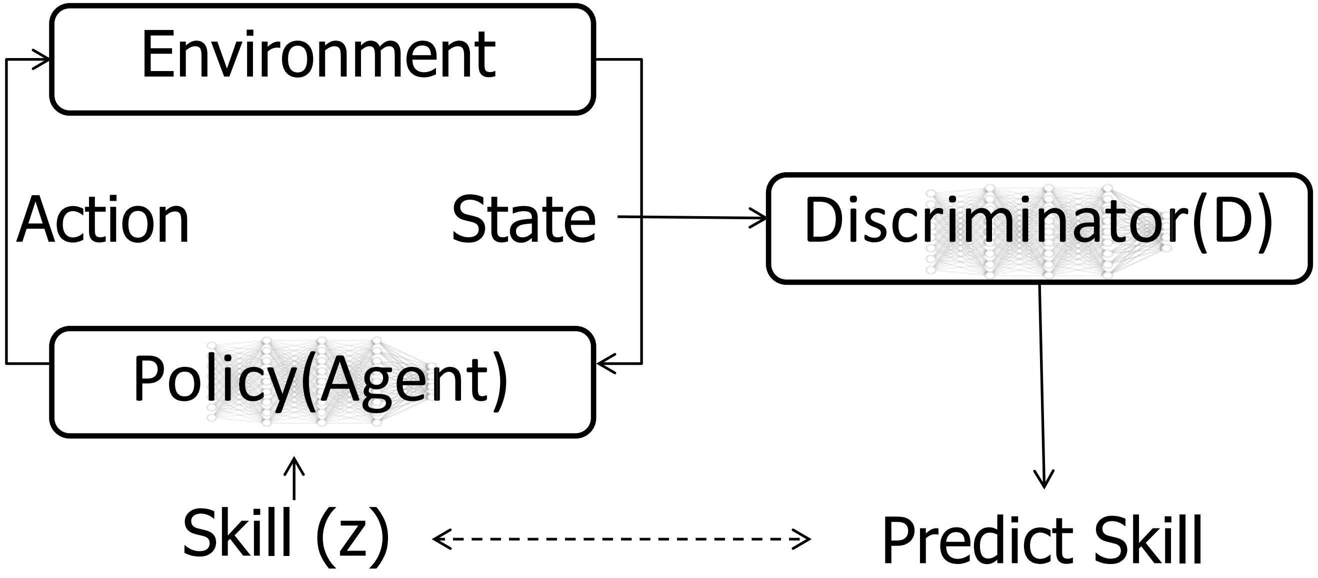
$$\pi(\mathbf{a}|\mathbf{s}, z) = \arg\max_{\pi} \sum_{z} E_{\mathbf{s}\sim\pi(\mathbf{s}|z)}[r(\mathbf{s}, z)]$$

reward states that are unlikely for other $z' \neq z$



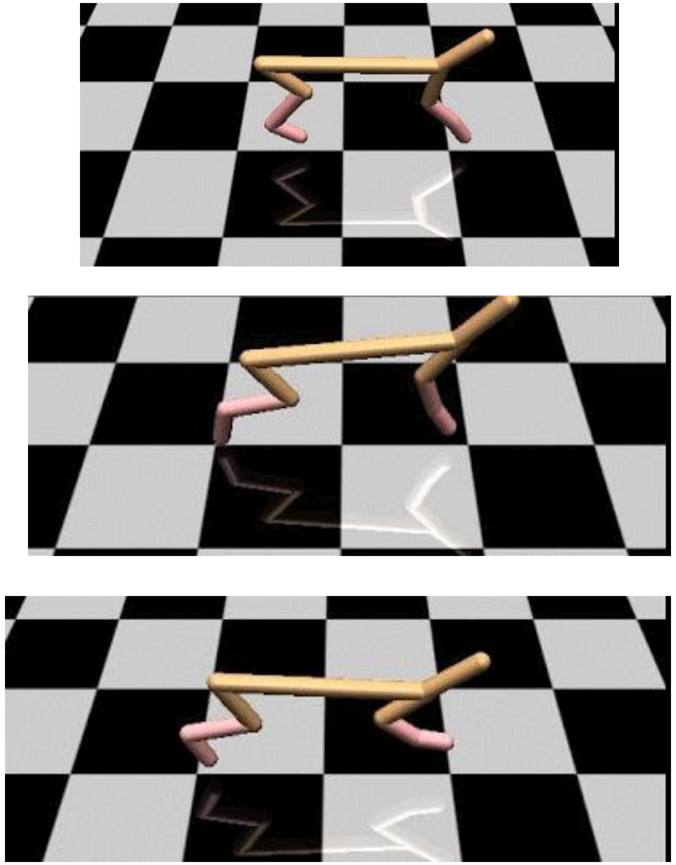
$$r(\mathbf{s}, z) = \log p(z|\mathbf{s})$$



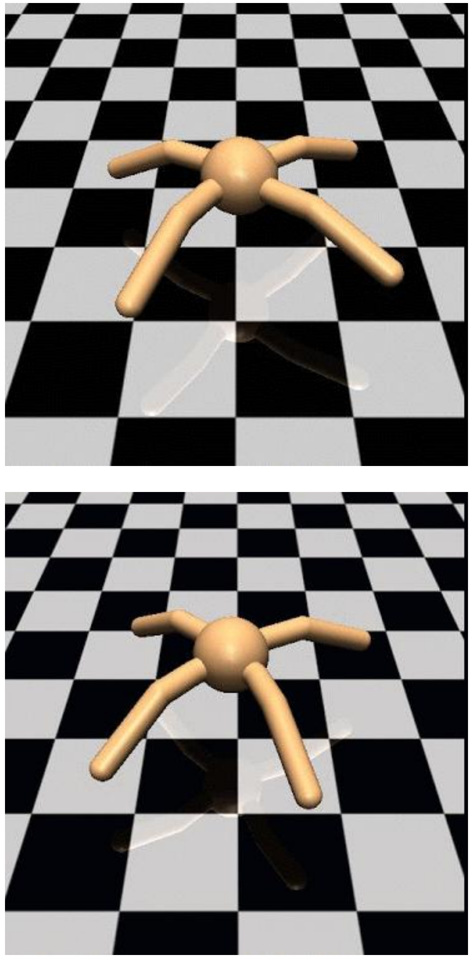


Eysenbach, Gupta, Ibarz, Levine. **Diversity is All You Need.**

Examples of learned tasks



Cheetah



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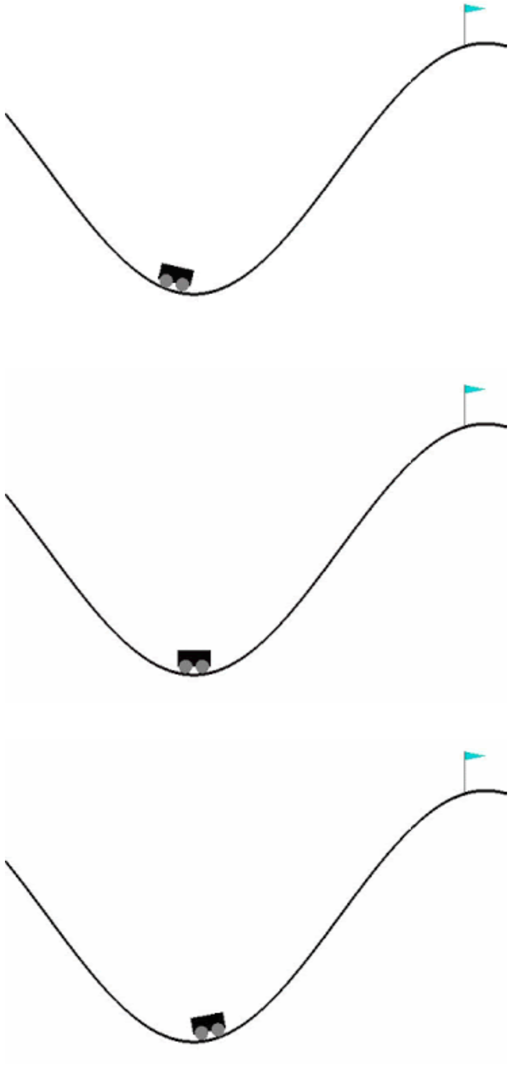


Mountain car

Eysenbach, Gupta, Ibarz, Levine. **Diversity is All You Need.**

A connection to mutual information

$$\pi(\mathbf{a}|\mathbf{s}, z) = \arg\max_{\pi} \sum_{z} E_{\mathbf{s}\sim\pi(\mathbf{s}|z)}[r(\mathbf{s}, z)]$$

$$r(\mathbf{s}, z) = \log p(z|\mathbf{s})$$

$$I(z;\mathbf{s}) = H(z) - H(z|\mathbf{s})$$

maximized by using uniform prior $p(z)$

minimized by maximizing $\log p(z|\mathbf{s})$

Eysenbach, Gupta, Ibarz, Levine. **Diversity is All You Need.**

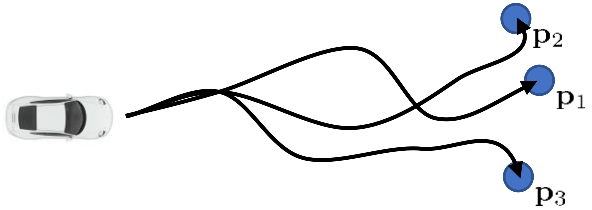
See also: Gregor et al. **Variational Intrinsic Control.** 2016

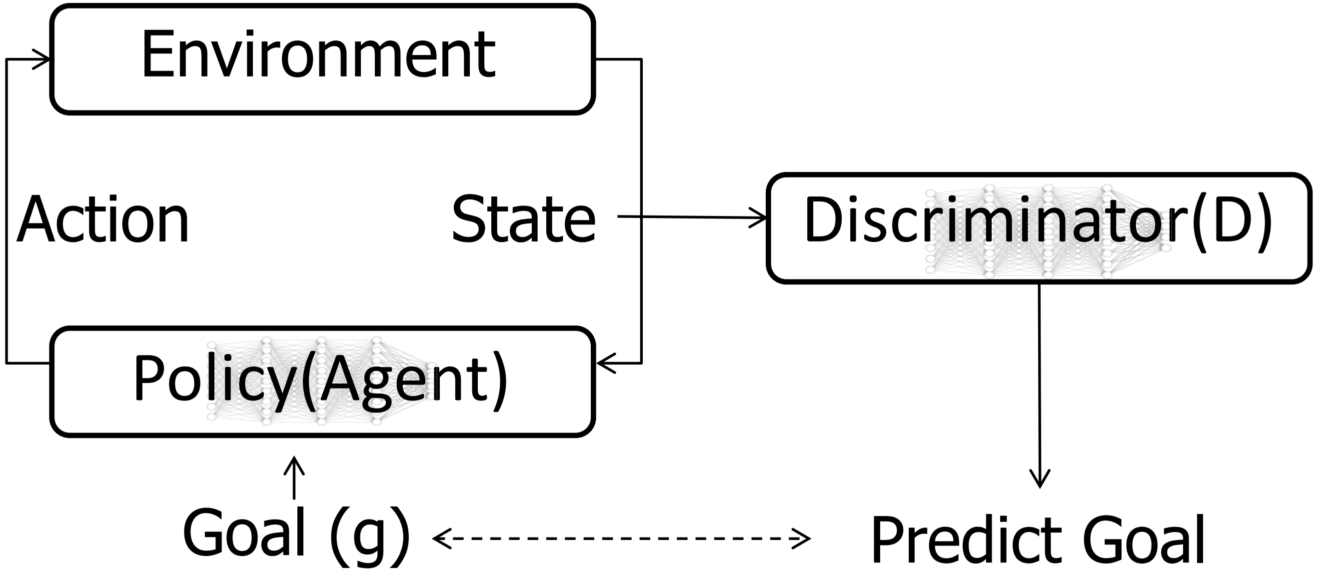
Part 3:
Reaching goals with empowerment

A little tweak

$$\pi(\mathbf{a}|\mathbf{s},\mathbf{g}) = \arg\max_{\pi} \sum_{z} E_{\mathbf{s}\sim\pi(\mathbf{s}|\mathbf{g})}[r(\mathbf{s},\mathbf{g})]$$

$$r(\mathbf{s},\mathbf{g}) = \log p(\mathbf{g}|\mathbf{s})$$





Goal reaching as MI maximization

$$\pi(\mathbf{a}|\mathbf{s},\mathbf{g}) = \arg\max_{\pi} \sum_{z} E_{\mathbf{s}\sim\pi(\mathbf{s}|\mathbf{g})}[r(\mathbf{s},\mathbf{g})]$$

$$r(\mathbf{s},\mathbf{g}) = \log p(\mathbf{g}|\mathbf{s})$$

in practice, we often use other rewards

$$r(\mathbf{s},\mathbf{g}) = \delta(\mathbf{s}=\mathbf{g})$$

$$r(\mathbf{s},\mathbf{g}) = \delta(\|\mathbf{s}-\mathbf{g}\| \leq \epsilon)$$

$$I(\mathbf{g};\mathbf{s}) = H(\mathbf{g}) - H(\mathbf{g}|\mathbf{s})$$

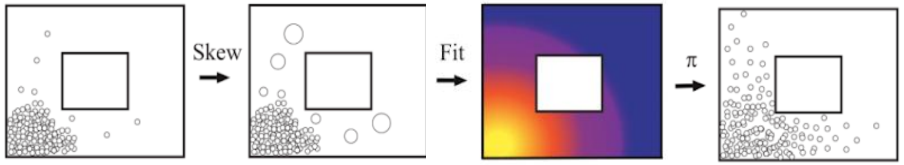
maximized by... proposing diverse goals?

minimized by maximizing $\log p(\mathbf{g}|\mathbf{s})$

this is often where a lot of interesting stuff happens

Example: Skew-Fit

1. Propose goal: $\mathbf{g} \sim p_\psi(\mathbf{g})$
2. Attempt to reach goal using $\pi(\mathbf{a}|\mathbf{s}, \mathbf{g})$
3. Use data to update π
4. Use data to update $p_\psi(\mathbf{g})$



standard MLE: $\psi \leftarrow \arg\max_\psi E[\log p_\psi(\mathbf{g})]$

weighted MLE: $\psi \leftarrow \arg\max_\psi E[w(\mathbf{g}) \log p(\mathbf{g})]$

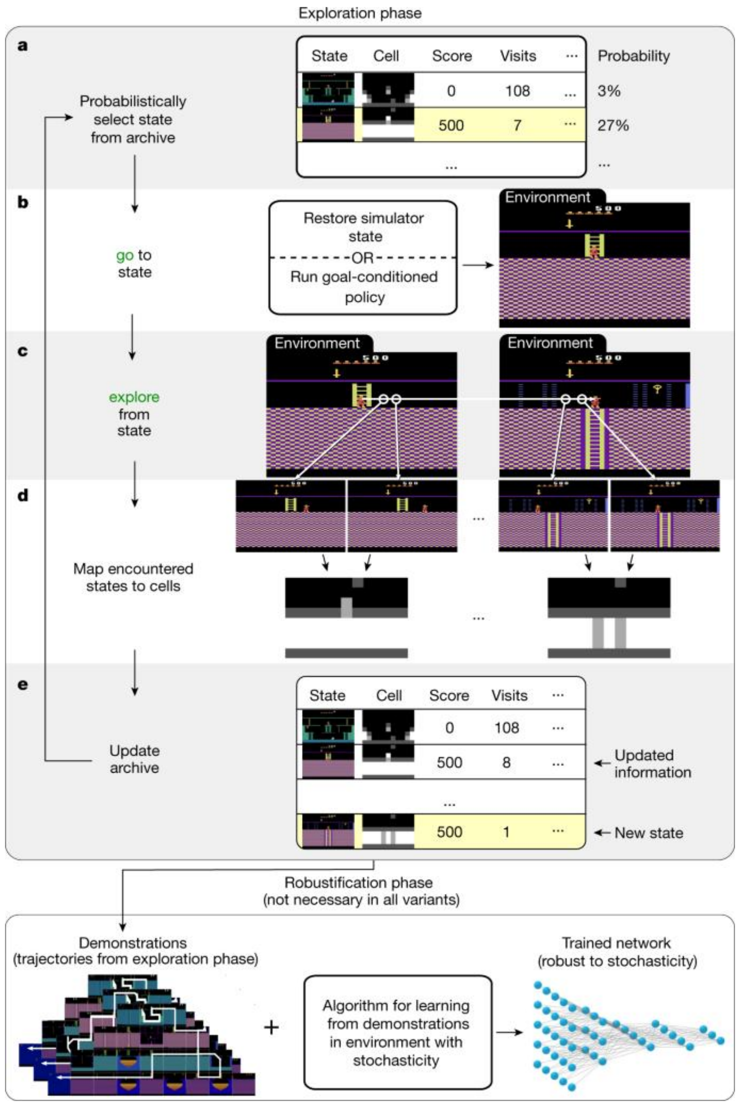
$$w(\mathbf{g}) = p_\psi(\mathbf{g})^\alpha$$

key result: for any $\alpha \in [-1, 0)$, entropy $\mathcal{H}(p_\psi(\mathbf{g}))$ increases!

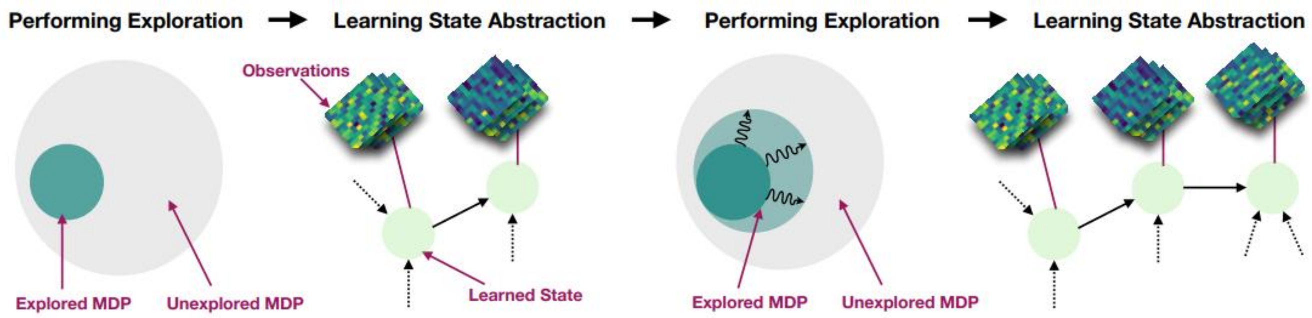
what might be suboptimal about this method?

Nair*, Pong*, Bahl, Dalal, Lin, L. **Visual Reinforcement Learning with Imagined Goals**. ’18
Dalal*, Pong*, Lin*, Nair, Bahl, Levine. **Skew-Fit: State-Covering Self-Supervised Reinforcement Learning.** ‘19

Lots of other ways to explore with goals!



Ecoffet et al., Go-explore



Misra et al., Homer (Kinematic State Abstraction)